



sdi:1.24

API Reference

X Platform, SDI JPEG XS Encoder TS 2.26.2

28/08/2025

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1 Overview

Changelog:

1.24

- **Changed**
 - portstatus from 1.19 to 1.20

1.23

- **Added**
 - CardSettings version 1.0
- **Changed**
 - physicalports from 1.16 to 1.17

1.22

- **Changed**
 - portstatus from 1.18 to 1.19

1.21

- **Changed**
 - portstatus from 1.17 to 1.18

1.20

- **Changed**
 - videoFormats from 1.0 to 1.1
 - physicalports from 1.15 to 1.16
 - outputstatus from 1.16 to 1.17
 - portstatus from 1.16 to 1.17
 - jpegxsoutputstatus from 1.6 to 1.7

1.19

- **Added**
 - sdiUtils version 1.0
- **Changed**
 - physicalports from 1.14 to 1.15

1.18

- **Changed**
 - physicalports from 1.13 to 1.14
 - outputstatus from 1.14 to 1.15
 - portstatus from 1.14 to 1.15
 - jpegxsoutputstatus from 1.4 to 1.5

1.17

- **Changed**

- physicalports from 1.12 to 1.13
- outputstatus from 1.13 to 1.14
- portstatus from 1.13 to 1.14
- jpegxsoutputstatus from 1.3 to 1.4

1.16

- **Added**

- inputRedundancy version 1.0

- **Changed**

- physicalports from 1.11 to 1.12
- outputstatus from 1.12 to 1.13
- portstatus from 1.12 to 1.13
- jpegxsoutputstatus from 1.2 to 1.3

1.15

- **Changed**

- outputstatus from 1.11 to 1.12

1.14

- **Added**

- component version 1.4
- tsstatus version 1.3

- **Changed**

- physicalports from 1.10 to 1.11
- outputstatus from 1.10 to 1.11
- portstatus from 1.11 to 1.12
- jpegxsoutputstatus from 1.1 to 1.2
- cardinfo from 1.1 to 1.2

1.13

- **Changed**

- physicalports from 1.9 to 1.10
- outputstatus from 1.9 to 1.10
- portstatus from 1.10 to 1.11
- jpegxsoutputstatus from 1.0 to 1.1
- transceiverStatus from 1.0 to 1.1

2 baseUuidNamespace (1.0)

3 cardinfo (1.2)

3.1 Overview

Changelog:

1.2

- **Added**
 - The field `hasTsEncapsulation` was added to `CardInfo` to support TS encapsulation.
- **Changed**
 - The field `isTicoXS` in `CardInfo` was renamed to `isJpegXS`

1.1

- **Added**
 - The field `isTicoXS` was added to `CardInfo` to support Tico XS.

3.2 Command Reference

3.2.1 GetCardInfo

- message `GetCardInfo.Request`
- message `GetCardInfo.Response`

3.3 Type Reference

3.3.1 CardInfo

struct

<code>displayName</code>	<code>string</code>
<code>isOptical</code>	<code>bool</code>
<code>hasOutputsOnly</code>	<code>bool</code>
<code>hasInputsOnly</code>	<code>bool</code>
<code>supportsPassthrough</code>	<code>bool</code>
<code>isTicoHD</code>	<code>bool</code>
<code>isTicoUHD</code>	<code>bool</code>
<code>isJpegXS</code>	<code>bool</code>
<code>hasTsEncapsulation</code>	<code>bool</code>

3.3.2 CardInfoRequest

empty **struct**

3.3.3 CardInfoResponse

struct

<code>data</code>	<code>CardInfo</code>
-------------------	-----------------------

3.3.4 GetCardInfo.Request

CardInfoRequest

3.3.5 GetCardInfo.Response

CardInfoResponse

4 cardsettings (1.0)

4.1 Overview

Changelog:

1.0

- Added
 - The CardSettings object was added.
 - The TimecodeSource enum was added.
 - The GetCardSettings and SetCardSettings RPCs were added.

4.2 Command Reference

4.2.1 GetCardSettings

RPCs for getting and setting the card settings.

- message **GetCardSettings.Request**
- message **GetCardSettings.Response**

4.2.2 SetCardSettings

- message **SetCardSettings.Request**
- message **SetCardSettings.Response**

4.3 Type Reference

4.3.1 CardSettings

SDI Card-wide settings.

struct

timecodeSource

TimecodeSource

Timecode source used for the whole SDI card.

4.3.2 GetCardSettings.Request

empty **struct**

4.3.3 GetCardSettings.Response

struct

data

CardSettings

4.3.4 SetCardSettings.Request

struct

data

CardSettings**4.3.5 SetCardSettings.Response**empty **struct****4.3.6 TimecodeSource**

Timecode source used for the whole SDI card.

enum

PTP	Timecode source is PTP.
NTP	Timecode source is NTP.

5 component (1.4)

5.1 Overview

Changelog:

1.4

- Added
 - str2ComponentType mapping, the inverse of componentType2str.
- Changed
 - Added DVBTMLSUB and JPEG_XS_VIDEO to ComponentType type.

5.2 Type Reference

5.2.1 AnyComponent

empty **struct**

5.2.2 ComponentGroup

ComponentGroup: Grouping of ComponentType, used for filtering

enum

UNDEFINED
 AUDIO
 VIDEO
 SUBTITLE
 DATA

5.2.3 ComponentLanguage

string

5.2.4 ComponentSelector

Component selector.

@param pid	Selects a component using a specific (source) PID.
@param language	Selects a component by a three character language string.
@param type	Selects a component by its type.
@param group	Selects a component by group membership.
@param any	Selects any component.

variant

pid	Pid
language	ComponentLanguage
componentType	ComponentType
componentGroup	ComponentGroup
any	AnyComponent

5.2.5 ComponentType

A ComponentType is typically identified by a stream type (see ISO 13818-1 Table 2-29), often distinguished with another header field such as a descriptor tag.

enum

MPEG_VIDEO	
MPEG_AUDIO	
TTX	
DVBSUB	
ECM	
PCR	
PRIVATE	
AC_3	
H264	
VBI	
AAC	
VC_1	
SCS_EMM	
DPI_CUE	
EAC3	
AIT	
HBBTV_CAROUSEL	
DATA_CAROUSEL	
H265	
S302M	
JPEG2000	
S2038	
DVBTTMLSUB	
JPEG_XS_VIDEO	

5.2.6 Pid

int

5.2.7 SubtitleFormat

enum

DVB	
DVB_HARD_OF_HEARING	
EBU	
EBU_HARD_OF_HEARING	

6 firewallTypes (1.0)

6.1 Type Reference

6.1.1 IPAddress

string

6.1.2 PaginatedRequest

Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request

@param pageNumber what page of the set to returned, determined by the slice [pageSize*pageNumber

struct

pageSize	int
pageNumber	int

6.1.3 PaginationInfo

struct

totalEntries	int
--------------	-----

the total number of entries on the server, before matching

6.1.4 SocketAddress

Tuple with IP address and port

struct

address	IPAddress
	IP address, either IPv4 or IPv6
port	int

6.1.5 UUID

string

7 inputRedundancy (1.0)

7.1 Type Reference

7.1.1 IdenticalSources

struct

bufferDelay	int Delay in milliseconds for offset handling between streams.
guardInterval	int Delay in milliseconds used to compensate for fluctuations in the time offset between the flows
latencyTuning	LatencyTuning Selected latency tuning strategy

7.1.2 LatencyTuning

Enum describing desired latency tuning strategy

enum

Auto	The first flow activated will be assumed to be leading
MainLeading	The Main source is assumed to be leading
MainLagging	The Main source is assumed to be lagging

7.1.3 SliplessSwitching

struct

identicalSources	optional IdenticalSources When the sources are expected to be identical, we allow more settings to improve latency performance
switchOnTei	bool When enabled, FPGA module additionally uses TEI status to assess alternative health. Available only for slipless fpga switching. Only one alternative allowed.

8 jpegxoutputstatus (1.7)

8.1 Overview

Changelog:

1.7

- **Changed**
 - bump videoFormats version from 1.0 to 1.1.

1.6

- **Changed**
 - Using videoFormats version 1.0 instead of physicalports module to get the Format type-def.

1.5

- **Changed**
 - bump physicalports version from 1.13 to 1.14.

1.4

- **Changed**
 - bump physicalports version from 1.12 to 1.13.

1.3

- **Changed**
 - bump physicalports version from 1.11 to 1.12.

1.2

- **Added**
 - import physicalports version 1.11 has been added.
 - import tsstatus version 1.3 has been added.
 - JpegXSNetworkDelayStatus structure has been added.
 - XsTsOutputStatus structure has been added.
 - Added rpc GetJpegXSNetworkDelayStatus.
 - Added rpc GetTsStatus.
 - Added rpc GetXsTsOutputStatus.
 - Added Protocol GetJpegXSNetworkDelayStatusProtocol.
 - Added Protocol GetTsStatusProtocol.
 - Added Protocol XsTsOutputStatus.

1.1

- **Added**
 - The structure JpegXSOffsetStatus was added.
 - Added RPC GetJpegXSOffsetStatus.
 - Added Protocol OffsetStatus.

- **Changed**

- The field `JpegXsAudioStatus` in `JpegXSAudioStatus` function description was renamed to `JpegXSAudioStatus`.

8.2 Command Reference

8.2.1 GetJpegXSAudioStatus

RPCs

- message `GetJpegXSAudioStatus.Request`
- message `GetJpegXSAudioStatus.Response`

8.2.2 GetJpegXSOffsetStatus

- message `GetJpegXSOffsetStatus.Request`
- message `GetJpegXSOffsetStatus.Response`

8.2.3 GetJpegXSNetworkDelayStatus

- message `GetJpegXSNetworkDelayStatus.Request`
- message `GetJpegXSNetworkDelayStatus.Response`

8.2.4 GetTsStatus

- message `GetTsStatus.Request`
- message `GetTsStatus.Response`

8.2.5 GetXsTsOutputStatus

- message `GetXsTsOutputStatus.Request`
- message `GetXsTsOutputStatus.Response`

8.3 Type Reference

8.3.1 GetJpegXSAudioStatus.Request

empty `struct`

8.3.2 GetJpegXSAudioStatus.Response

`struct`

data `map` from `int` to `JpegXSAudioStatus`

8.3.3 GetJpegXSNetworkDelayStatus.Request

empty `struct`

8.3.4 GetJpegXSNetworkDelayStatus.Response

struct

data **map** from **int** to **JpegXSNetworkDelayStatus**

8.3.5 GetJpegXSOffsetStatus.Request

empty **struct**

8.3.6 GetJpegXSOffsetStatus.Response

struct

data **map** from **int** to **JpegXSOffsetStatus**

8.3.7 GetTsStatus.Request

empty **struct**

8.3.8 GetTsStatus.Response

struct

data **map** from **UUID** to **tsstatus.TransportStreamStatus**

8.3.9 GetXsTsOutputStatus.Request

empty **struct**

8.3.10 GetXsTsOutputStatus.Response

struct

data **map** from **int** to **XsTsOutputStatus**

8.3.11 JpegXSAudioStatus

JpegXSAudioStatus

struct

CH1Sync	optional bool Optional bool, if not nullopt the audio channel is active, if true, it is also synced.
CH2Sync	optional bool
CH3Sync	optional bool
CH4Sync	optional bool
CH5Sync	optional bool
CH6Sync	optional bool
CH7Sync	optional bool

CH8Sync	optional bool
CH9Sync	optional bool
CH10Sync	optional bool
CH11Sync	optional bool
CH12Sync	optional bool
CH13Sync	optional bool
CH14Sync	optional bool
CH15Sync	optional bool
CH16Sync	optional bool

8.3.12 JpegXSNetworkDelayStatus

JpegXSNetworkDelayStatus

struct

ptpActive	bool Is true if ptp is active on the box.
video	int The video essence network delay.
anc	int The Ancillary essence newtwork delay.
audios	map from int to int A map of port number to audio essence network delays.

8.3.13 JpegXSOffsetStatus

JpegXSOffsetStatus

struct

video	int The video essence delay.
anc	int The Ancillary essence delay.
audios	map from int to int A map of port number to audio essence delays.

8.3.14 XsTsOutputStatus

struct

videoMode	videoFormats.Format
videoFractional	bool
videoUHD	bool
imageWidth	int
imageHeight	int
frameRate	int
syncLossCount	int

9 outputstatus (1.17)

9.1 Overview

Changelog:

1.17

- **Changed**
 - bump videoFormats version from 1.0 to 1.1

1.16

- **Changed**
 - PhysicalPorts is no longer a dependency of OutputStatus. videoFormats.1.0.idcl is used instead to get the Format typedef.

1.15

- **Changed**
 - bump physicalports version from 1.13 to 1.14.

1.14

- **Changed**
 - bump physicalports version from 1.12 to 1.13.

1.13

- **Added**
 - The fields effectiveIdenticalSourceOffset, mainBufferDelay, backupBufferDelay were added to OutputSourceStatus to support Hot Input Redundancy.
- **Changed**
 - bump physicalports version from 1.11 to 1.12.

1.12

- **Added**
 - Added AlternativeStatus structure.
 - Added OutputSourceStatus structure.
 - The field outputSources was added to SdiOutputStatus.

1.11

- **Changed**
 - bump physicalports version from 1.10 to 1.11.

1.10

- **Changed**
 - bump physicalports version from 1.9 to 1.10.

1.9

- **Changed**
 - bump physicalports version from 1.8 to 1.9.

1.8

- **Changed**
 - bump physicalports version from 1.7 to 1.8.

1.7

- **Changed**
 - bump physicalports version from 1.6 to 1.7.

1.6

- **Changed**
 - bump physicalports version from 1.5 to 1.6.

1.5

- **Changed**
 - bump physicalports version from 1.4 to 1.5.

1.4

- **Changed**
 - bump physicalports version from 1.3 to 1.4.

1.3

- **Changed**
 - bump physicalports version from 1.2 to 1.3.

1.2

- **Changed**
 - bump physicalports version from 1.1 to 1.2.

1.1

- **Changed**
 - bump physicalports version from 1.0 to 1.1.

9.2 Command Reference

9.2.1 GetSdiOutputStatus

RPCs

- message **GetSdiOutputStatus.Request**
- message **GetSdiOutputStatus.Response**

9.3 Type Reference

9.3.1 AlternativeStatus

struct

isBroken **bool**

9.3.2 ClockFrequency

enum

NO_TIME_STAMP	
CF_27_MHZ	
CF_1485_MHZ	
CF_1485_1001_MHZ	
CF_297_MHZ	
CF_297_1001_MHZ	

9.3.3 Fec

enum

NO_FEC	
L_FEC	
L_ADN_D_FEC	

9.3.4 Frame

enum

FRAME_480I	
FRAME_576I	
FRAME_1080I	
FRAME_1080P	
FRAME_1080I_PSF	
FRAME_2KP	
FRAME_2KI_PSF	
FRAME_720P	
UNKNOWN	

9.3.5 Frate

enum

FR_2970_GHZ	
FR_2970_1001_GHZ	
FR_1485_GHZ	
FR_1485_1001_GHZ	
FR_0270_GHZ	
FR_60	
FR_60_1001	
FR_50	
FR_48	
FR_48_1001	
FR_30	
FR_30_1001	
FR_25	

FR_24
FR_24_1001
UNKNOWN

9.3.6 GetSdiOutputStatus.Request

empty **struct**

9.3.7 GetSdiOutputStatus.Response

struct

data **map** from **UUID** to **SdiOutputStatus**

9.3.8 Map

enum

DIRECT_SAMPLE
SMPTE_B_DL
SMPTE_B_DS

9.3.9 OutputSourceStatus

struct

effectiveIdenticalSourceOffset	optional int
mainBufferDelay	optional int
backupBufferDelay	optional int
currentAlternativeIdx	int
alternativeStatuses	list of AlternativeStatus

9.3.10 ReferenceTime

enum

NOT_LOCKED
RESERVED
LOCKED.UTC
LOCKED.PRIVATE

9.3.11 Sample

enum

UNKNOWN
S_422_10
S_444_10
S_4444_10

RESERVED

S_422_12

S_444_12

S_4444_12

S_4224_12

9.3.12 SdiOutputStatus

struct

videoMode	videoFormats.Format
missingPktCount	int
syncLossCount	int
extension	int
formatPresent	bool
vsid	Vsid
frameCount	int
refTime	ReferenceTime
scrambled	bool
fec	Fec
cf	ClockFrequency
streamStructure	Map
frame	Frame
frate	Frate
sample	Sample
outputSources	list of OutputSourceStatus

9.3.13 Vsid

enum

PRIMARY

PROTECT

10 physicalports (1.17)

10.1 Overview

Changelog:

1.17

- **Added**
 - Added the field `timecodeInsertion` to `TsAncSettings`.

1.16

- **Changed**
 - `videoFormats` from 1.0 to 1.1

1.15

- **Changed**
 - The UUID of the `PhysicalPort` object is now deterministic instead of random. See the `sdi-Tools.1.0` module for how to calculate this deterministically.
 - The `Format` typedef was moved to `common/videoFormats.1.0` module to reduce amount of modules needing to be bumped whenever the `physicalports` module is changed.

1.14

- **Added**
 - Added `CHECKFIELD` to `Pattern` enum that can be used to generate a pathological pattern.

1.13

- **Added**
 - The enum `PortCapacity` was added to specifies the maximum format capacity of the port.
 - The field `portCapacity` was added to the `PhysicalPort` structure.
 - The rpc `DeletePhysicalPorts` was added to support deleting settings for one or more physical ports.

1.12

- **Added**
 - The import `inputRedundancy` version 1.0 was added.
 - The field `sliplessSwitching` were added to the `InputRedundancy` structure.

1.11

- **Added**
 - The structure `TsAudioSettings` was added to support audio configuration for inputs on a JPEG XS TS product.
 - The structure `TsVideoSettings` was added to support video configuration for inputs on a JPEG XS TS product.
 - The structure `TsAncSettings` was added to support ancillary configuration for inputs on a JPEG XS TS product.
 - The structure `TsServiceSettings` was added to support service configuration for inputs on a JPEG XS TS product.

- The structure `TsSettings` was added to support source configuration for inputs on a JPEG XS TS product.
- The enum `FlowType` was added to support Flow types.
- The fields `flowType` and `tsSettings` were added to `InputSettings` structure.
- The structure `TsDecAudio` was added to support information used to define each audio on the jpeg xs ts decoder.
- The structure `TsDecAudioSettings` was added to define all audios on a jpeg xs ts decoder card.
- The fields `flowType` and `tsDecAudioSettings` were added to `OutputSettings` structure.

1.10

- **Added**
 - The enum `FrameSync` was added to specify which reference signal we want to synchronize our video output to.
 - The field `frameSync` was added to `PhysicalPort` structure .
 - The fields `chStart` and `chWidth` were added to `ST2110Audio` structure.
- **Removed**
 - The field `channel` was removed from `ST2110Audio` structure .

1.9

- **Added**
 - The value `CUSTOM` was added to enum `TicoVideoCompressionRatio` .
 - The field `bitrate` was added to `TicoVideoEncodeConfiguration` structure.
 - The rpc `ClearAllCounters` was added.
 - The protocol `ClearAllCountersProtocol` was added.

1.8

- **Added**
 - The value `TEST_GENERATOR` was added to enum `Pattern` .
 - The enum `InputRedundancySwitchMode` was added to specify the switchMode between alternative sources.
 - The structure `PreferredSource` was added to specify preferred source configuration.
 - The structure `InputRedundancy` was added to specify data structure related to Input Redundancy feature.
 - The field `inputRedundancy` was added to `OutputSettings` structure.
- **Changed**
 - The enum `Format` was updated with ASI value and names of the VM1080lxx values.
 - The enum `AudioChannelRangeSelection` was updated with new values.

1.7

- **Added**
 - The enum `TextPosition` was added to specify the vertical placement for the text.
 - The structure `Color` was added to support color defined in terms of RGB.

- The enum `BitmapBackgroundProperty` was added to specifies the color property of the background in a bitmap.
- The structure `TextOverlay` was added to support text overlay to put on top of a test image.
- The structure `MovingBox` was added to support moving box to be used in a test image generator.
- The structure `TestGenerator` was added to defines how to generate a test image for a specific port.
- The enum `TrafficPolicing` was added to specifies the different modes that are available for the trafficPolicing.
- The field `testGenerator` and `trafficPolicing` were added to the `PhysicalPort` structure.
- **Changed**
 - The enum was renamed from `ExpectedFormat` to `Format` and updated with formats `VM480P` and `VM576P`.
 - The enum was renamed from `NoSignalPattern` to `Pattern`.

1.6

- **Added**
 - The field `bufferDelay` from `ST2110GroupSource` structure was added.
 - The structure `ST2110Audio` was added.
- **Changed**
 - The field `anc` from `ST2110GroupSource` structure was changed from `uuid` to `optional(uuid)` type.
 - The field `audio` from `ST2110GroupSource` structure was changed from `list(optional(uuid))` to `list(ST2110Audio)` type.

1.5

- **Added**
 - The enum `NoSignalPattern` was added.
 - The structure `ST2110GroupSource` was added.
 - The field `noSignalPattern` was added to the `PhysicalPort` structure.
 - The field `groupSource` was added to the `OutputSettings` structure.

1.4

- **Changed**
 - The enumeration `PortDirection` has been updated with `Quad` values.
- **Removed**
 - The protocol `SetPhysicalPortProtocol` and `DeletePhysicalPortProtocol` were removed.

1.3

- **Added**
 - The protocol `SetPhysicalPortProtocol` and `DeletePhysicalPortProtocol` were added.
- **Changed**
 - The enumeration `AudioChannelRangeSelection` has been updated.

1.2

- **Added**

- The enumeration `AudioInterval` was added to support time between audio packets.
- The enumeration `AudioEncapsulation` was added to support encapsulation format of the encoded audio channel.
- The enumeration `AudioChannelRangeSelection` was added to specifies input channels.
- The structure `AudioMapping` was added to support encoded audio channel.
- The enumeration `TicoVideoCompressionRatio` was added to specifies compression strength of the encoded video given in ratios.
- The enumeration `TicoVideoOptimization` was added to specifies what the encoded video should be optimized for.
- The structure `TicoVideoEncodeConfiguration` was added to specifies options for the TICO encoder.
- The fields `ticoConfiguration` and `audioMappings` were added to the `InputSettings`.

- **Changed**

- The type of the input field in `PortSettings` variant was change from `optional(InputSettings)` to `InputSettings`
- The type of the output field in `PortSettings` variant was change from `optional(OutputSettings)` to `OutputSettings`

1.1

- **Added**

- The field `id` was added to the `PhysicalPort`.

10.2 Command Reference

10.2.1 GetPhysicalPorts

RPC for retrieving information regarding all ports, or only specific one if a filter is specified.

- message `GetPhysicalPorts.Request`
- message `GetPhysicalPorts.Response`

10.2.2 SetPhysicalPorts

RPC for submitting settings for one or more physical ports.

- message `SetPhysicalPorts.Request`
- message `SetPhysicalPorts.Response`

10.2.3 DeletePhysicalPorts

RPC for deleting settings for one or more physical ports. Should never be used.

- message `DeletePhysicalPorts.Request`
- message `DeletePhysicalPorts.Response`

10.2.4 ClearAllCounters

- message `ClearAllCounters.Request`

- message **ClearAllCounters.Response**

10.3 Type Reference

10.3.1 AudioChannelRangeSelection

Specifies which input channels should be used.

enum

e1_2	Use channels 1 and 2.
e3_4	Use channels 3 and 4.
e5_6	Use channels 5 and 6.
e7_8	Use channels 7 and 8.
e9_10	Use channels 9 and 10.
e11_12	Use channels 11 and 12.
e13_14	Use channels 13 and 14.
e15_16	Use channels 15 and 16.
e1_4	
e5_8	
e9_12	
e13_16	
e1_6	Use channels 1, 2, 3, 4, 5 and 6.
e2_7	Use channels 2, 3, 4, 5, 6 and 7.
e3_8	Use channels 3, 4, 5, 6, 7 and 8.
e4_9	Use channels 4, 5, 6, 7, 8 and 9.
e5_10	Use channels 5, 6, 7, 8, 9 and 10.
e6_11	Use channels 6, 7, 8, 9, 10 and 11.
e7_12	Use channels 7, 8, 9, 10, 11 and 12.
e8_13	Use channels 8, 9, 10, 11, 12 and 13.
e9_14	Use channels 9, 10, 11, 12, 13 and 14.
e10_15	Use channels 10, 11, 12, 13, 14 and 15.
e11_16	Use channels 11, 12, 13, 14, 15 and 16.
e1_8	
e9_16	

10.3.2 AudioEncapsulation

Encapsulation format of the encoded audio channel.

enum

SMPTE_2110_30	AES67 audio transport format.
SMPTE_2110_31	AES3 audio transport format.

10.3.3 AudioInterval

Time between audio packets.

enum

e1_MS	1 millisecond
-------	---------------

e125_US	125 nanoseconds
---------	-----------------

10.3.4 AudioMapping

Definition of an encoded audio channel created from a selection of incoming audio channels.

struct

interval	AudioInterval The rate between audio packets.
encapsulation	AudioEncapsulation The encapsulation format to utilize.
channelSelection	AudioChannelRangeSelection The incoming audio channels to encode.

10.3.5 BitmapBackgroundProperty

Specifies the color property of the background in a bitmap.

enum

BLACK	Make the bitmap background solid black.
TRANSPARENT	Make the bitmap background transparent, such that the bitmap in essence becomes an overlay.

10.3.6 ClearAllCounters.Request

empty **struct**

10.3.7 ClearAllCounters.Response

empty **struct**

10.3.8 Color

A color defined in terms of RGB.

struct

red	int Amount of red. Value 0 to 255.
green	int Amount of green. Value 0 to 255.
blue	int Amount of blue. Value 0 to 255.

10.3.9 DeletePhysicalPorts.Request

Message to send to card to delete one or more physical ports.

struct

ids

list of UUID

A list of physical port UUIDs that should be deleted.

10.3.10 DeletePhysicalPorts.Response

A simple acknowledgement message to indicate success. If the request is faulty then an error is returned instead.

empty **struct****10.3.11 FlowType****enum**

UNCOMPRESSED_RTP_2022_6

TICO_UHD_2022_6

JPEG_XS_2110

JPEG_XS_TS

10.3.12 FrameSync

Containing information used to synchronize frames.

struct

frameSyncMode

FrameSyncMode

mode used to synchronize frames.

genlockOffset

optional int

optional field for setting pulse delay, in case genlock is used to sync frames on JPEG XS 2110 decoder.

10.3.13 FrameSyncMode

Specifies which reference signal we want to synchronize our video output to

enum

DISABLED

Frame sync disabled

GENLOCK

Frame sync to genlock

PTP

Frame sync to PTP

10.3.14 GetPhysicalPorts.Request

Request either all physical ports with an empty request, or only specific physical ports matching the specified filter.

variant

filter

GetPortFilter

The filter to utilize when fetching physical ports.

10.3.15 GetPhysicalPorts.Response

Contains the requested physical ports.

struct

data	map from UUID to PhysicalPort A map of ports returned by the request.
info	firewallTypes.PaginationInfo Meta information about the data - i.e the total number of items on the server

10.3.16 GetPortFilter

Specifies a search for specific ports.

struct

pagination	optional firewallTypes.PaginatedRequest Settings for splitting the response data into pages
filter	optional PortFilter Search term used to filter the data before pagination

10.3.17 InputRedundancy

Data structure related to Input Redundancy feature

struct

mode	InputRedundancySwitchMode Selected switch mode
backups	list of UUID List of alternative sources
preferredSource	optional PreferredSource When set allows to override default source preferences
switchbackDelay	int Delay in seconds to wait before switching back to a more preferred alternative or MAIN source.
sliplessSwitching	inputRedundancy.SliplessSwitching The FPGA will handle the redundancy switching without glitches, given identical sources.

10.3.18 InputRedundancySwitchMode

Enum related to SwitchMode between alternative sources

enum

Disabled	InputRedundancy is inactive; only MAIN source is used.
Reverting	Automatically selects first healthy alternative source including MAIN source; i.e. no alarms detected on the source

10.3.19 InputSettings

Settings specific to an input port.

struct

inputId	UUID Unique identifier for the input.
flowType	FlowType Type of the input port configured, depending on the product.
ticoConfiguration	optional TicoVideoEncodeConfiguration If not set no TICO encoding will take place. Otherwise this represents the options for the encoder.
audioMappings	list of optional AudioMapping Specifies how the 16 incoming audio channels should be grouped and encoded. Each individual incoming audio channel can be used in at most 1 mapping. The list always has a size of 8 with unused audio mappings being set to None.
tsSettings	optional TsSettings Used only if flowType=JPEG_TS, will be an empty optional on others. Specifies the TS source information for the input. If this is used, audioMappings field above should be set to empty list since TS related audio is configured inside this type.

10.3.20 MovingBox

Specify the properties of a moving box to be used in a test image generator.

struct

enabled	bool Include this moving box in the test image to be generated.
color	optional Color The Color (R, G, B) of the box. Default is orange.

10.3.21 OutputFormat

Specifies the format of the output SDI signal.

enum

SD	Standard Definition output.
HD	High Definition output, 720p or higher.

10.3.22 OutputSettings

Settings specific to an output port.

struct

outputId	UUID The unique identifier for the output.
	UUID

source	The unique identifier for the source that will be used as output.
groupSource	optional ST2110GroupSource If set, the source is a ST2110 Group Source, and is identified with this UUID.
nextSource	optional UUID •
sources	map from UUID to Service A list of all the services that make up the output.
format	OutputFormat Specifies the format of the output.
inputRedundancy	InputRedundancy
tsDecAudioSettings	optional TsDecAudioSettings Specifies the audio settings of the jpeg xs ts decoder.
flowType	FlowType Type of the input port configured, depending on the product.

10.3.23 Pattern

Specifies an image pattern.

enum

NONE	Do not generate any image. Not available for TestGenerator.
BLACK_PICTURE	Generate an entirely black frame.
COLOR_BAR	Generate an image consisting of several distinctive, vertical color bars.
FREEZE_FRAME	Freeze on the last valid frame seen on the relevant port. Not available for Test Generator.
TEST_GENERATOR	Enable Test Generator when no signal detected.
CHECKFIELD	Generate a pathological image.

10.3.24 PhysicalPort

Specifies all relevant settings relating to a single physical SDI port.

struct

enabled	bool Enables the port for input/output. If disabled, then the port is unusable.
portNumber	int The number of the port. If it is electrical, then it is the logical, physical ordering (1,2,3,4,...) if optical, then the first port has a value of 4, and the second port has a value of 1.
label	string The user-modifiable nickname for this port, in order to easily distinguish it from the rest.
	videoFormats.Format

expectedFormat	The expected format of the SDI video this input/output port is assuming to receive. If there is a large discrepancy between the expected and actual format, then the traffic policer will kick in and mute the input/output.
portSettings	PortSettings Defines direction-specific (input/output) port settings.
direction	PortDirection Specifies the port direction. A port can either be an input or an output, not both.
noSignalPattern	Pattern Specifies the image to be generated when there is a signal loss.
testGenerator	TestGenerator Specifies the test image generator, which can be useful for initial signal testing. The test generator image can also be used as a noSignalPattern image if specified.
trafficPolicing	TrafficPolicing Specifies a maximum rate for the input signal. If the set rate is exceeded the FPGA mutes the stream.
frameSync	FrameSync Containing information used to synchronize frames
portCapacity	optional PortCapacity Specifies the maximum format capacity of the port. Only used for XS TS Enc optical, which is why it's optional.

10.3.25 PortCapacity

Specifies the maximum format capacity of the port

enum

NO	The port will not handle anything. In essence it is turned off.
HD	The port can handle up to HD formats/bitrates
UHD	The port can handle up to UHD formats/bitrates

10.3.26 PortDirection

Specifies the direction of the port.

enum

INPUT	The port is to receive an ingoing SDI signal.
OUTPUT	The port is to produce an outgoing SDI signal.
PASSTHROUGH	The port is to pass a signal directly to another port on the same card.
CLONED	The port is one of two or more ports on the card that sends an identical video.
QUAD_INPUT	For UHD coding. Groups four HD signals (a quadrant each) into one UHD video.
QUAD_OUTPUT_SQUARE_DIVISION	For UHD coding. Takes a UHD signal and splits it into four quadrants (four HD signals), going clockwise from the upper-left quadrant.

QUAD_OUTPUT_INTERLEAVED

For UHD coding. Uses 2-sample interleave. Groups of two pixels are distributed across four links, Even rows (further divided into two-pixel groups) are sent across link 1 and 2, odd rows are sent on link 3 and 4.

10.3.27 PortFilter

Specifies a filter to be used as part of GetPhysicalPort RPCs and similar.

struct

searchTerm **optional string**

10.3.28 PortSettings

A physical port can be either input or output type port.

variant

input

InputSettings

If PortSettings equal an InputSettings type, then this indicates that it is an input port.

output

OutputSettings

If PortSettings equal an OutputSettings type, then this indicates that it is an output port.

10.3.29 PreferredSource

Preferred source configuration

struct

index

int

Selects a source that should be marked as preferred. Index = 0 is MAIN source, index = 1 means first backup, etc. Preferred source will be used as the default source when it is healthy. If it fails the system will switch using regular source priorities (for example: first MAIN, then first backup, then second backup and so on)

locked

bool

When enabled disables switching to another source when preferred source is faulty

10.3.30 ST2110Audio

Defines a set of audio packets that are used as part of a ST2110 stream.

struct

essence

UUID

Uniquely identifies the audio.

chStart

int

Identifies to which channel map first audio channel from essence

chWidth	int Identifies how many channels is contained in the essence
---------	--

10.3.31 ST2110GroupSource

Describes a logical grouping of ST2110 sources.

struct

bufferDelay	int Maximum allowable delay in milliseconds.
isTico	bool If flag is set, then the video is TICO-encoded.
video	UUID Unique identifier for the video source.
anc	optional UUID Unique identifier for ancillary data packets.
audio	list of ST2110Audio Defines the audio packets that are part of this group source.

10.3.32 Service

Service information

struct

name	string The name of the service.
sid	int The service ID.
thumbnail	string The corresponding thumbnail.

10.3.33 SetPhysicalPorts.Request

Message to send to card to set one or more physical port settings.

struct

data	map from UUID to PhysicalPort A map of physical port UUID and the corresponding Physical-Port object to set.
------	--

10.3.34 SetPhysicalPorts.Response

A simple acknowledgement message to indicate success. If the request is faulty then an error is returned instead.

empty **struct**

10.3.35 TestGenerator

Defines how to generate a test image for a specific port.

struct

force	bool Immediately override the input/output signal on the port and display the test image with desired settings. If false, the test generated image will only display if it is set as the noSignal-Pattern on the PhysicalPort, and there is a signal loss.
format	videoFormats.Format The format of the test image. Can be up to fullHD resolutions on inputs. Outputs can have resolutions up to ultraHD.
pattern	Pattern The TestGeneratorPattern to generate as base image.
textOverlay	TextOverlay If flag is set, then the port label will be displayed on the test generated image.
movingBox	MovingBox If set, a box with the desired color will be re-rendered on the test-image at a set interval to confirm that the signal has not frozen.
audio	bool If enabled, a constant audio tone will be generated alongside the test-generated image in order to confirm a continuous, non-interrupted audio signal.

10.3.36 TextOverlay

Specify the properties of a text overlay to put on top of a test image.

struct

enabled	bool Enable the text overlay. The label of the port will be used as the text. If the text is too long for the image, it will be shortened.
position	TextPosition Specify where to place the text on the vertical plane. It will always be centered horizontally.
background	BitmapBackgroundProperty Specify the properties of the background in the bitmap.

10.3.37 TextPosition

Specify the vertical placement for the text.

enum

TOP	Place it at the top of the image.
MIDDLE	Place it in the middle of the image.
BOTTOM	Place it at the bottom of the image.

10.3.38 TicoVideoCompressionRatio

Compression strength of the encoded video given in ratios.

enum

e1_2	Compress to a ratio of 1:2.
e1_4	Compress to a ratio of 1:4.
e1_5	Compress to a ratio of 1:5.
CUSTOM	

10.3.39 TicoVideoEncodeConfiguration

Specifies options for the TICO encoder.

struct

compressionRatio	TicoVideoCompressionRatio The compression strength to utilize.
optimization	TicoVideoOptimization Specifies what the encoded video should be optimized for.
bitrate	optional float

10.3.40 TicoVideoOptimization

Specifies what the encoded video should be optimized for.

enum

PSNR	Optimize for Peak Signal-to-Noise Ratio.
VISUAL	Optimize for perceived visual quality.

10.3.41 TrafficPolicing

Specifies the different modes that are available for the trafficPolicing

enum

TP_ZERO	Nothing shall pass through the policer
TP_SD	Everything up to SD will pass through
TP_HD	Everything up to HD will pass through
TP_3G	Policer will allow everything

10.3.42 TsAncSettings

Ancillary configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

struct

ancPid	int Ancillary pid for the service.
	bool

timecodeInsertion

Whether or not to enable timecode insertion in the ancillary data. The timecode source used will be the timecode source specified in CardSettings.

10.3.43 TsAudioSettings

Audio configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

struct

pid

int

Pid number of this audio channel in resulting TS stream.

channelSelection

AudioChannelRangeSelection

The incoming audio channels to encode.

10.3.44 TsDecAudio

Containing information used to define each audio on the jpeg xs ts decoder.

struct

pid

int

number of this audio channel in resulting TS stream. When autoSelectPids is true, this value will be ignored.

chStart

int

Identifies which channel the audio mapping will start on.

chWidth

int

Identifies how many channels will be contained in the audio.

10.3.45 TsDecAudioSettings

Settings to define all audios on a jpeg xs ts decoder card.

struct

autoSelectPids

bool

Defines which allocation method to apply to pids for the audios. True means the the lowest numbered pid will be the first audio selected, false means filling out the pids manually.

audioMappings

list of TsDecAudio

Defines a list of the audios.

10.3.46 TsServiceSettings

Service configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

struct

serviceName

string

Name of the service.

int

serviceId	ID number of the service.
pmtPid	int PMT pid for the service.
pcrPid	int PCR pid for the service.
pcrInterval	float Interval in which PCR is signalled repeatedly (in milliseconds).

10.3.47 TsSettings

Source configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

struct

tsServiceSettings	TsServiceSettings Service settings for the source.
tsVideoSettings	TsVideoSettings Video settings for the source.
tsAncSettings	TsAncSettings Ancillary settings for the source.
tsAudioSettings	list of optional TsAudioSettings Audio settings for the source.

10.3.48 TsVideoSettings

Video configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

struct

videoPid	int Video pid for the service.
----------	--

11 portstatus (1.20)

11.1 Overview

Changelog:

1.20

- **Added**
 - The rpc GetSfpStatus was added to get SFP status for the ports.
 - Imported the sfpStatus version 1.5 module.

1.19

- **Changed**
 - Changed the format variable in AudioChannelStatus from AudioFormat to optional(AudioFormat)

1.18

- **Added**
 - AesMode, AesChannelMode, and analysedMaxLevel fields added to AudioChannelStatus

1.17

- **Changed**
 - videoFormats from 1.0 to 1.1

1.16

- **Changed**
 - Using videoFormats version 1.0 instead of physicalports module to get the Format type-def.

1.15

- **Changed**
 - physicalports from 1.13 to 1.14

1.14

- **Changed**
 - physicalports from 1.12 to 1.13

1.13

- **Changed**
 - physicalports from 1.11 to 1.12

1.12

- **Changed**
 - physicalports from 1.10 to 1.11

1.11

- **Changed**
 - physicalports from 1.9 to 1.10
 - transceiverStatus from 1.0 to 1.1

1.10

- **Changed**
 - physicalports from 1.8 to 1.9

11.2 Command Reference

11.2.1 GetSfpStatus

Get SFP Status.

Get status and diagnostics information reported from the SFP plug for applicable ports (only applicable for optical ports).

- message **GetSfpStatus.Request**
- message **GetSfpStatus.Response**

11.2.2 GetPortStatus

- message **GetPortStatus.Request**
- message **GetPortStatus.Response**

11.3 Type Reference

11.3.1 AesChannelMode

enum

MONO
STEREO
MULTI
N_A
UNKNOWN

11.3.2 AesMode

enum

PCM
NON_PCM

11.3.3 AudioChannelStatus

Status of an incoming audio channel.

struct

format	optional AudioFormat
aesMode	AesMode
aesChannelMode	AesChannelMode
analysedMaxLevel	optional float
samplingRate	AudioSamplingRate

11.3.4 AudioFormat

Audio format of an incoming audio channel.

enum

PCM

DOLBY_DIGITAL

DOLBY_DIGITAL_PLUS

DOLBY_E

11.3.5 AudioSamplingRate

Sampling rate in kHz.

enum

SYNC_48

ASYNC_48

SYNC_44_1

ASYNC_44_1

SYNC_32

ASYNC_32

SYNC_FREE_RUNNING

ASYNC_FREE_RUNNING

11.3.6 GetPortStatus.Request

variant

ids

list of UUID

11.3.7 GetPortStatus.Response

struct

data

map from UUID to PortStatus

11.3.8 GetSfpStatus.Request

GetSfpStatusRequest

11.3.9 GetSfpStatus.Response

GetSfpStatusResponse

11.3.10 GetSfpStatusRequest

Get SFP Status Request

struct

physicalports

optional list of UUID

List of UUIDs uniquely identifying PhysicalPorts. Only SFP Status from the specified PhysicalPorts are returned. If no value is set, the request will return SFP Status from all ports.

11.3.11 GetSfpStatusResponse

struct

data

map from UUID to optional **sfpStatus.SfpStatus**

11.3.12 PortStatus

PortStatus

struct

lock

bool

inputFormat

videoFormats.Format

syncLostCount

int

edhErrorCount

int

audioStatuses

list of optional **AudioChannelStatus**

A list of length 16 corresponding to the statuses of the 16 audio channels. If no audio is detected is on the channel the status will be None.

transceiverStatus

optional **transceiverStatus.TransceiverStatus**

12 quadinputstatus (1.0)

12.1 Command Reference

12.1.1 GetQuadInputStatus

- message **GetQuadInputStatus.Request**
- message **GetQuadInputStatus.Response**

12.2 Type Reference

12.2.1 GetQuadInputStatus.Request

empty **struct**

12.2.2 GetQuadInputStatus.Response

struct

data **map** from **int** to **QuadInputStatus**

12.2.3 QuadInputMode

enum

UNKNOWN
SQUARE_DIVISION
INTERLEAVED

12.2.4 QuadInputStatus

struct

instance	int Ports 1-4 instance 0, 5-8 instance 1 ...
hasLock	bool
lockLossCount	int
mode	QuadInputMode Mode recived from fpga
ancMode	QuadInputMode Mode recived in ANC frame for this instance
skew	list of float Skew for channel in ns

13 sdiTools (1.0)

14 sfpConstants (1.4)

14.1 Overview

Changelog

1.4

- **Changed**
- Changed "SFP or SFP+" to "SFP/SFP+/SFP28" The corresponding identifier refers to the given 3 SFP types as per SNIA SFF-8024 specification.

14.2 Type Reference

14.2.1 Identifier

Physical Device Identifier Values. (SFF 8472 Rev12.3 spec Table 5-1)

enum

UNKNOWN	Unknown or unspecified
GBIC	GBIC
SOLDERED	Module soldered to motherboard (ex. SFF)
SFP_SFPP	SFP, SFP+ or SFP28
QSFP	QSFP+
QSFP28	QSFP28
NOT_USED	Not used by SFF 8472 Rev12.2.1. These values are maintained in the Transceiver Management section of SFF-8024.
VENDOR	Vendor specific

14.2.2 Transceiver

Transceiver Compliance Codes. (SFF 8472 Rev12.3 spec Table 5-3)

enum

ER_10G	10G Base-ER
LRM_10G	10G Base-LRM
LR_10G	10G Base-LR
SR_10G	10G Base-SR
ACTIVE_CABLE_40G	40G Active cable
LR4_40G	40G Base-LR4
SR4_40G	40G Base-SR4
CR4_40G	40G Base-CR4
PX	BASE-PX
BX10	BASE-BX10
FX_100	100BASE-FX
LX_100	100BASE-LX/LX10
T_1000	1000BASE-T
CX_1000	1000BASE-CX
LX_1000	1000BASE-LX
SX_1000	1000BASE-SX

ELECTRICAL_INTER_ENCLOSURE	Electrical inter enclosure
LONGWAVE_LASER_LL	Longwave laser LL
MEDIUM	Medium distance
LONG_DISTANCE	Long distance
INTERMEDIATE_DISTANCE	Intermediate distance
SHORT_DISTANCE	Short distance
VERY_LONG_DISTANCE	Very long distance
ACTIVE_CABLE	Active Cable
PASSIVE_CABLE	Passive Cable
LONGWAVE_LASER_LC	Longwave laser LC
SHORTWAVE_LASER_w_OFC	Shortwave laser w OFC
SHORTWAVE_LASER_wo_OFC	Shortwave laser w/o OFC
ELECTRICAL_INTRA_ENCLOSURE	Electrical intra enclosure
SINGLE_MODE	Singel mode
MULTIMODE_50um	Multimode 50um
MULTIMODE_50m	Multimode 50m
MULTIMODE_625m	Multimode 62.6m
VIDEO_COAX	Video coax
MINIATURE_COAX	Miniature coax
SHIELDED_TWISTED_PAIR	Shielded twisted pair
TWIN_AXIAL_PAIR	Twin axial pair
M_BYTES_SEC_100	100M bytes per sec
M_BYTES_SEC_200	200M bytes per sec
M_BYTES_SEC_400	400M bytes per sec
M_BYTES_SEC_1600	1600M bytes per sec
M_BYTES_SEC_800	800M bytes per sec
M_BYTES_SEC_1200	1200M bytes per sec
AOC_100G_AOC_25GAUI_C2M_BER_5_	Extended Specification Compliance. (SFF-8024 Rev 4.8 spec Table 4-4)
SR4_100G_BASE_SR_25G_BASE	100GBASE-SR4 or 25GBASE-SR
LR4_100G_BASE_LR_25G_BASE	100GBASE-LR4 or 25GBASE-LR
ER4_100G_BASE_ER_25G_BASE	100GBASE-ER4 or 25GBASE-ER
SR10_100G_BASE	100GBASE-SR10
CWDM4_100G	100G CWDM4
PSM4_PARALLEL_SMF_100G	100G PSM4 Parallel SMF
ACC_100G_ACC_25GAUI_C2M_BER_5_	100G ACC (Active Copper Cable) or 25GAUI C2M ACC. Providing a worst BER of $5 * 10^{-5}$
CR4_100G_BASE_25G_BASE_CR_CA_2	100GBASE-CR4 or 25GBASE-CR CA-L
CR_25G_BASE_CA_25G_S_50G_BASE_	25GBASE-CR CA-S
CR_25G_BASE_CA_25G_N_50G_BASE_	25GBASE-CR CA-N
ER4_40G_BASE	40GBASE-ER4
SR_4_X_10G_BASE	4 x 10GBASE-SR
PSM4_PARALLEL_SMF_40G	40G PSM4 Parallel SMF
G959_1_PROFILE_P1I1_2D1	G959.1 profile P1I1-2D1 (10709 MBd, 2km, 1310nm SM)
G959_1_PROFILE_P1S1_2D2	G959.1 profile P1S1-2D2 (10709 MBd, 40km, 1550nm SM)
G959_1_PROFILE_P1L1_2D2	G959.1 profile P1L1-2D2 (10709 MBd, 80km, 1550nm SM)
T_10G_BASE	10GBASE-T with SFI electrical interface

CLR4_100G	100G CLR4
AOC_100G_AOC_25GAUI_C2M_BER_10	100G AOC or 25GAUI C2M AOC. Providing a worst BER of 10^{-12} or below
ACC_100G_ACC_25GAUI_C2M_BER_10	100G ACC or 25GAUI C2M ACC. Providing a worst BER of 10^{-12} or below
DWDM2_100GE	100GE-DWDM2 (DWDM transceiver using 2 wavelengths on a 1550nm DWDM grid with a reach up to 80km)

15 sfpStatus (1.5)

15.1 Overview

Changelog

1.5

- **Changed**
 - sfpConstants from 1.3 to 1.4

15.2 Type Reference

15.2.1 SfpDiagnostics

Real Time Diagnostic for SFP/SFP+/QSFP28/QSFP+, with A2h/A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 9-11) (SFF 8636 Rev2.10 spec Table 6-1)

struct

temp	float Internally measured module temperature (°C). [96-97], [22-23]
vcc	float Internally measured supply voltage in transceiver (V). [98-99], [26-27]
txPwr	list of float Measured TX output power (mW). [102-103], [50-57]
rxPwr	list of float Measured RX input power (mW). [104-105], [34-41]

15.2.2 SfpStatus

SFP Status Data Fields, with A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 4-1) QSFP A0h memory space address is specified next to the SFP A0h. (SFF 8636 Rev 2.10 spec Table 6-14)

struct

identifier	sfpConstants.Identifier Type of transceiver (SFF 8024 Rev4.8a spec Table 4-1) [0],[128]
transceiver	list of sfpConstants.Transceiver Code for electronic or optical compatibility [3-10], [131-138]
brNominal	float Nominal signalling rate, units of 100 MBd. [12], [140/222]
vendorName	string SFP vendor name. [20-35], [148-163]
vendorOUI	string SFP vendor IEEE company ID. [37-39], [165-167]
vendorPN	string Part number provided by SFP vendor. [40-55], [168-183]
	string

vendorRev	Revision level for part number provided by vendor. [56-59], [184-185]
wavelength	string Laser wavelength. [60-61], [186-187]
vendorSN	string Serial number provided by SFP vendor. [68-83], [196-211]
dateCode	string Vendor's manufacturing date code. [84-91], [212-219]
diagnostics	optional SfpDiagnostics Optional Diagnostics Monitor Data.

16 ticooutputstatus (1.1)

16.1 Command Reference

16.1.1 GetTicoOutputStatus

- message **GetTicoOutputStatus.Request**
- message **GetTicoOutputStatus.Response**

16.2 Type Reference

16.2.1 AudioInputInfo

enum

NO_AUDIO_DATA	
AI_16CH_LINK_1	
AI_16CH_LINK_2	
AI_16CH_LINK_3	
AI_16CH_LINK_4	
AI_32CH_LINK_1	
AI_32CH_LINK_2	
AI_32CH_LINK_3	
AI_32CH_LINK_4	
AI_16CH_LINK_1_PLUS_16CH_LINK_	
AI_16CH_LINK_1_PLUS_16CH_LINK_	
AI_16CH_LINK_1_PLUS_16CH_LINK_	
AI_16CH_LINK_2_PLUS_16CH_LINK_	
AI_16CH_LINK_2_PLUS_16CH_LINK_	
AI_16CH_LINK_3_PLUS_16CH_LINK_	

16.2.2 ColorPrimaries

enum

UNSPECIFIED	
ITU_R_BT601_7_D65	
ITU_R_BT709_6_D65	
ITU_R_BT2020_2_D65	
SMPTE_RP_431_2_DCI	
SMPTE_RP_431_2_D65	
SRGB_D65	

16.2.3 FrameRate

enum

NO_DEFINED_VALUE	
RESERVED	
FR_23_976	

FR_24
FR_48_1001
FR_25
FR_29_97
FR_30
FR_48
FR_50
FR_59_64
FR_60
FR_100
FR_120_1001
FR_120

16.2.4 GetTicoOutputStatus.Request

empty **struct**

16.2.5 GetTicoOutputStatus.Response

struct

data **map** from **UUID** to **TicoOutputStatus**

16.2.6 InterlaceMode

enum

PROGRESSIVE
INTERLACED_FIRST_FIELD
INTERLACED_SECOND_FIELD

16.2.7 MatrixCoefficient

enum

UNSPECIFIED
ITU_R_BT709_6
ITU_R_BT601
ITU_R_BT2020_2

16.2.8 SampleBitdepth

enum

SB_8_BITS
SB_10_BITS
SB_12_BITS

16.2.9 SamplingStructure

enum

SS_422_YCBCR	
SS_444_YCBCR	
SS_444_GBR	
SS_420	

16.2.10 SourceType

enum

UNKNOWN	
SDI	
NATIVE	

16.2.11 TicoOutputStatus

struct

instanceNumber	int
ticoDetected	bool
bitstreamProfile	int
frameRate	FrameRate
interlaceMode	InterlaceMode
sampleBitdepth	SampleBitdepth
samplingStructure	SamplingStructure
horizontalSize	int
verticalSize	int
colorPrimaries	ColorPrimaries
transferCharacter	TransferCharacter
matrixCoefficient	MatrixCoefficient
sourceType	SourceType
videoInputInfo	VideoInputInfo
audioInputInfo	AudioInputInfo
vbiInputInfo	VbiInputInfo
maxFifoFilling	int

16.2.12 TransferCharacter

enum

UNSPECIFIED	
ITU_R_BT601_BT709_6_BT2020_2	
SMPTE_ST_2084_2014	
SLOG_3	
ARIB_STD_B67	

16.2.13 VbiInputInfo

enum

NO_VBI_DATA

VBI_LINK_1

VBI_LINK_2

VBI_LINK_3

VBI_LINK_4

16.2.14 VideoInputInfo

enum

UNKNOWN

VI_QUAD_SQD_LEVEL_A

VI_QUAD_SQD_LEVEL_B

VI_QUAD_2SI_LEVEL_A

VI_QUAD_2SI_LEVEL_B

VI_12G_SDI

VI_6G_SDI

VI_DUAL_3G_SDI

VI_QUAD_HD_SDI

17 transceiverStatus (1.1)

17.1 Type Reference

17.1.1 CableLengthIndicator

Information about the cable length indicator measurement. The measurement is given in terms of the Belden 1694A cable.

struct

length	int An indication of the cable length carrying the SDI input signal [m].
lengthPrecision	int The precision of the cable length measurement [m]. The actual length of the cable is likely to be in the interval [length - precision, length + precision].
maxLength	int The maximum supported cable length [m].
threshold	int The threshold below which measurements of cable length are very imprecise [m].

17.1.2 DataRate

The various input data rates the transceiver can lock onto.

enum

DR_11_88_GBPS	
DR_5_94_GBPS	
DR_2_97_GBPS	
DR_1_485_GBPS	
DR_270_MBPS	
NO_LOCK	

17.1.3 SignalProperties

struct

dataRate	DataRate An indication of the data rate of the SDI signal that the transceiver has locked.
cableLength	optional CableLengthIndicator An indication of the cable length carrying the SDI input signal [m]. N/A if dataRate is NO_LOCK.
horizontalEyeOpening	optional float Horizontal eye opening of the equalized SDI signal [UI]. Not available if dataRate is NO_LOCK. Not available for SD.

17.1.4 TransceiverStatus

Status about the quality of an SDI signal. These measurements are taken by the transceiver which is connected to the physical connector carrying the SDI signal.

struct

carrierDetect	bool True if the transceiver has detected that there is an input signal from an SDI cable.
signalProperties	optional SignalProperties Properties of the input signal that are available if a carrier is detected.

18 tsstatus (1.3)

18.1 Overview

Changelog:

1.3

- Changed
 - Dependency module component bumped from version 1.1 to 1.4

18.2 Type Reference

18.2.1 OutputTransportStreamStatus

Status for the output TS

struct

services	map from int to ServiceSourceStatus Status for each service in the TS. Map from service ID to status struct.
pids	list of PidStatus Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.

18.2.2 PidStatus

PidStatus state of PID

struct

serviceId	list of int List of serviceIds a PID belong to.
componentType	optional component.ComponentType ComponentType of PID
ccErrors	int CC errors on PID
dataRate	bigint Data rate of PID
scrambling	Scrambling Scrambling of PID (none, even or odd)
PCR	bool
present	bool True if signalled pid has rate
language	optional string Language of PID, if applicable (e.g. audio pid)
casId	optional int Id of CAS system, if applicable (e.g. ECM pid)
pidIn	int PID number on input
	optional int

pidOut	PID number on output
--------	----------------------

18.2.3 Scrambling

Scrambling state of PID

enum

NONE	No scrambling
EVEN	Scrambling with even control word
ODD	Scrambling with odd control word

18.2.4 ServiceSourceStatus

Status for an input service

struct

serviceName	optional string Service name
pmtPid	int PMT pid for service
pcrPid	optional int PCR pid for service if present

18.2.5 TransportStreamStatus

Status for the TS

struct

ccErrors	int Total CC error count for this TS.
services	map from int to ServiceSourceStatus Status for each service in the TS. Map from service ID to status struct.
pids	map from int to PidStatus Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.
tsId	optional int Transport Stream ID; set if PAT pid present in stream
originalNetworkId	optional int Original Network ID; set if SDT pid present in stream

19 videoFormats (1.1)

19.1 Overview

Changelog:

1.1

- Changed
 - Added video formats to Format: VM2160P25, VM2160P30, VM2160P30_1001

19.2 Type Reference

19.2.1 Format

Specifies the expected format of a SDI video signal on either the input or the output.

enum

OFF	This port is not expected to be used.
VM480I	(Video Mode) 480 horizontal lines, interlaced.
VM576I	576 horizontal lines, interlaced.
VM720P24	720 horizontal lines, progressive scan, 24 frames per second.
VM720P24_1001	720 horizontal lines, progressive scan, 24/1.001=23.97 frames per second (NTSC).
VM720P25	720 horizontal lines, progressive scan, 25 frames per second.
VM720P30	720 horizontal lines, progressive scan, 30 frames per second.
VM720P30_1001	720 horizontal lines, progressive scan, 30/1.001=29.97 frames per second (NTSC).
VM720P50	720 horizontal lines, progressive scan, 50 frames per second.
VM720P60	720 horizontal lines, progressive scan, 60 frames per second.
VM720P60_1001	720 horizontal lines, progressive scan, 60/1.001=59.94 frames per second (NTSC).
VM1080P24	1080 horizontal lines, progressive scan, 24 frames per second.
VM1080P24_1001	1080 horizontal lines, progressive scan, 24/1.001=23.97 frames per second (NTSC).
VM1080P25	1080 horizontal lines, progressive scan, 25 frames per second.
VM1080P30	1080 horizontal lines, progressive scan, 30 frames per second.
VM1080P30_1001	1080 horizontal lines, progressive scan, 30/1.001=29.97 frames per second (NTSC).
VM1080I25	1080 horizontal lines, interlaced, 25 frames per second.
VM1080I30	1080 horizontal lines, interlaced, 30 frames per second.
VM1080I30_1001	1080 horizontal lines, interlaced, 30/1.001=29.97 frames per second (NTSC).
VM1080P50	1080 horizontal lines, progressive scan, 50 frames per second.
VM1080P60	1080 horizontal lines, progressive scan, 60 frames per second.
VM1080P60_1001	1080 horizontal lines, progressive scan, 60/1.001=59.94 frames per second (NTSC).
VM2160P25	2160 horizontal lines, progressive scan, 25 frames per second.
VM2160P30	2160 horizontal lines, progressive scan, 30 frames per second.
VM2160P30_1001	2160 horizontal lines, progressive scan, 30/1.001=29.97 frames per second (NTSC).
VM2160P50	2160 horizontal lines, progressive scan, 50 frames per second.

VM2160P60	2160 horizontal lines, progressive scan, 60 frames per second.
VM2160P60_1001	2160 horizontal lines, progressive scan, $60/1.001=59.94$ frames per second (NTSC).
ASI	ASI signal over SDI.
UHD_HD_SD	Any format up to UHD.
HD_SD	Any format up to HD.